Gisma University of Applied Sciences
Department of Computer and Data Sciences

Module Handbook

M504 AI and Applications

Spring 2026

Module Summary

Code:	M504
Name:	AI and Applications
Level:	Master
Credits:	5 ECTS
Workload:	50 Hours Directed Study + 75 Hours Independent Study
Delievery Mode:	Hyflex
Prerequisites:	None
Corequisites:	None
Leader:	Prof. Dr. Mohammad Mahdavi
Tutor:	Dr. Sanjit Kumar Saha
Contact Info:	Sanjit.Kumar@gisma-education.com
Term:	Semester 2, Quarter 1

Description

Python is the primary programming language of data scientists. In this module, students will learn the foundation of Python programming together with its well-known data science libraries. Furthermore, students will learn to implement programs in practice using the Python programming language and its data science libraries.

Objectives

Specific objectives include

- To evaluate core concepts of Python programming for data science.
- To build data science pipelines with Python in action.

Learning Outcomes

On successful completion of the module, students will be able to

1. Critically evaluate the business contexts that can benefit from Python programming.
2. Critically analyze core Python programming concepts, including its core structures and data science libraries.
3. Critically build data science pipelines for various problems using Jupyter Notebooks, NumPy, and Pandas.

Teaching Methods

Students will engage in learning through a series of directed study activities, lectures, and self-study. The teaching sessions will be divided into two parts. The first part belongs to an interactive lecture, where students learn about theoretical aspects of the presented topic. The second part will be an interactive lab, where students will implement the presented concepts themselves under the guidance of the tutor. Students will be also directed to online materials for further study. Note that teaching sessions will not be recorded by default, but the tutor may decide to record certain parts of lectures from time to time.

Content Outline

Week	Topics	Learning Outcomes	References	Activities
1	Introduction to Python	All	Ch. 1, 2, and 3 [1]	Interactive Lecture; Practical Exercises
2	Expressions; Data Types	All	Ch. 5 and 6 [1]	Interactive Lecture; Practical Exercises
3	Control Statements	All	Ch. 4 [1]	Interactive Lecture; Practical Exercises; Summative Assessment
4	Functions	All	Ch. 7 [1]	Interactive Lecture; Practical Exercises
5	Classes	All	Ch. 9 and 10 [1]	Interactive Lecture; Practical Exercises; Summative Assessment
6	Modules and Packages; Code Style	All	Ch. 8 [1]; [2]	Interactive Lecture; Practical Exercises
7	NumPy	All	Ch. 2 [3]	Interactive Lecture; Practical Exercises; Summative Assessment
8	Pandas	All	Ch. 3 [3]	Interactive Lecture; Practical Exercises
9	Explanatory Data Analysis	All	Ch. 3 [3]	Interactive Lecture; Practical Exercises; Summative Assessment

References

[1] Vijay Kumar Sharma, Vimal Kumar, Swati Sharma, and Shashwat Pathak. *Python programming: A practical approach*. CRC Press, 2021.

[2] Guido Van Rossum, Barry Warsaw, and Nick Coghlan. Pep 8–style guide for python code. *Python.org*, 1565:28, 2001.

[3] Jake VanderPlas. *Python data science handbook: Essential tools for working with data*. “O’Reilly Media, Inc.”, 2016.